

L12 - Geometric Series

Thursday, April 16, 2020 7:41 AM

$$S = a + ar + ar^2 + ar^3 + \dots + ar^{n-2} + ar^{n-1}$$
$$rS = ar + ar^2 + ar^3 + \dots + ar^{n-2} + ar^{n-1} + ar^n$$

$$rS - S = ar^n - a$$
$$S(r - 1) = ar^n - a$$
$$S = (ar^n - a)/(r - 1) \quad (1)$$

Solving for

$$1 + 2 + 2^2 + 2^3 + \dots + 2^{n-2} + 2^{n-1}$$

Putting $a=1, r=2$ in equation 1

$$S = (2^n - 1)/(2 - 1) = 2^n - 1$$

For $r < 1$, we usually do the following

$$S - rS =$$

$$1/2^{k-1} + 1/2^{k-2} + \dots + 1/2^2 + 1/2 + 1$$

$$r=1/2$$

$$S = (1 * 1/2^k - 1)/(1/2 - 1)$$

$$= (1 * 1/2^k - 1)/(-1/2)$$

$$= -1/(-1/2) = 2$$